

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

EMT 528 — Past Questions & Materials
by Renance

Course: EMT 528
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 500

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Paper 1 of 1 — EMT 528 Miscellaneous Techniques in Environmental Analysis PQ

Level 500 • Second Semester • PastQuestion

DEPARTMENT OF ENVIRONMENTAL MANAGEMENT AND TOXICOLOG

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN ©

COLLEGE OF SCIENCE

SECOND SEMESTER 2022/2023 EXAMINATION

EMT 528; MISCELLANEOUS TECHNIQUES IN ENVIRONMENTAL ANALYSIS (2 UNITS)

TIME ALLOWED; 3:00 Hours

Instruction: Answer FOUR questions, at least one from each section.

Section

4

A calculator is allowed.

A

(a) Explain the fundamental principle behind X-ray Diffraction (XRD) and how it is used to analyse the crystalline structure of materials.

(b) The second-order diffraction peak was observed

wavelength 130pm. during an experiment performed on

spacing (d) of the crystal lattice, and express your answer in Angstroms.

X-ray Diffraction (XRD) as a technique for material

sampled at an angle of 28 degree with X-rays of

a crystal. Determine the interplanar

(a) Enumerate the advantages of using X-ray Diffraction

analysis

wavelength of 1.8 Angstroms was

(b) During a crystallography experiment, X-rays with a wavelength

of 1.8 Angstroms, what will be the

observed. If the crystal lattice has an interplanar spacing (d) of 1.5 Angstroms

at which the first-order diffraction peak is observed?

You are tasked with characterizing the crystal structure of a newly synthesized material using

X-ray Diffraction (XRD).

Discuss the step-by-step procedure you would follow, from sample

preparation to data analysis,

highlighting the importance of each stage in obtaining accurate

results.

Section B

4,

tas

- (a) Explain the fundamental principle behind Neutron Activation Analysis (NAA) and how it is used to detect pollutants in soil.
- (b) Provide a step-by-step description of the AA process, including the role of a neutron source and gamma-ray detection.
- (c) Mention the advantages of using NAA for soil pollution detection and provide one specific application of NAA in environmental assessments.

You are conducting a Neutron Activation Analysis (NAA) experiment to determine the concentration of lead (Pb) in a soil sample. During the experiment, you measure the number of gamma-ray counts (N) and the activity of the activated isotope (A). The irradiation time (t) is 600 seconds, and the mass of the soil sample (m) is 10 grams. Given that the gamma-ray detection efficiency (ϵ) is 0.75 and the decay constant (λ) is $2.21 \times 10^{-3} \text{ s}^{-1}$ for the activated isotope, calculate the concentration of lead (Pb) in the soil sample in parts per million (ppm). NOTE: [N=5000 counts; A=1000 decays per second (Becquerels)]

in an agricultural area, Explain how Neutron

Activation Analysis (NAA) can contribute to assessing the impact of agricultural practices on soil quality. Discuss the specific elements of interest and the potential pollutants that NAA can detect, and provide recommendations on how the results can be used for soil management and remediation.

Consider a soil pollution monitoring project

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load current is

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e. Back emf of motor is 300 V. Find the firing angle α of the converter.